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Development and application of safety analysis code for He-Xe cooled space reactor☆

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ABSTRACT

Space nuclear reactor power system can offer high energy density and a wide power coverage, providing electric power for space exploration. The safety characteristics of space nuclear reactor during in-orbit operation accidents is key to evaluate its feasibility. In this paper, a safety analysis code is developed for He-Xe mixture cooled space reactor based on SCTRAN, which used to be applied for Supercritical water or CO_2 cooled reactor. A semi empirical formula is applied to develop a physical property calculation model of He–Xe. Flow resistance and heat transfer correlations as well as turbomachinery models for He-Xe are developed. A numerical model for S^4-CBC space reactor power system is established by using the newly developed code. The calculation results of reactor and system model have the maximum relative errors of 3.43 % and 0.47 % compared to the reference design values, which proves the code is reliable to simulate the thermal–hydraulic behavior of the S^4-CBC space reactor power system. The transient responses of the S^4-CBC space reactor power system under the reactivity insertion accident and the mechanical failure of one and two Brayton loops are simulated and analyzed. The results of the accident analysis provide guidance for the core design, operation strategy, and safety system design of future He-Xe cooled space reactor system.

1. Introduction

Space power technology is the core of space exploration. Space nuclear reactor power system can offer high energy density and a wide power coverage, providing electric power for space exploration (Biao et al., 2020). It can also provide high power density energy for human migration on the moon or Mars. Energy conversion system in space reactors includes both static and dynamic system. Static system is characterized by low mass, simple structure, and high reliability, but it has low efficiency and power output, making its suitable for low-power tasks. Dynamic system has a more complex structure and lower reliability, but it can provide higher power and efficiency.

Dynamic system includes Rankine, Stirling, and Brayton system. Different task requirements for space reactor employ distinct dynamic systems, Table 1 shows the parameter of some space nuclear reactors with power range of kW – MW. The SNAP-SER (IECTE, 1984) reactor is a nuclear reactor system developed by America in the 1960 s for space missions, which has an electrical power of 5 kW and a thermal efficiency of 10 %. It employs NaK as the coolant and utilizes the Thermionic Energy Conversion System (TECS). The HOMER-15 (Poston, 2001) and AFSPS (Poston et al., 2008) which were developed by America in the 2000 s, are cooled by Na and NaK. It adopts the Stirling system with an electric power of 3 kW and a thermal efficiency of 20 %. The TEM (Hendrickx, 2019), MEGAHIT (Ruault et al., 2014) and Prismatic space reactor (Yang et al., 2017) are cooled by He-Xe, which adopt the Brayton system with an electric power over 1 MW. Therefore, Therefore, the Rankine system is susceptible to the effects of gravity, which complicates its use in microgravity environments. Water, the typical working fluid in Rankine systems, is particularly challenging for space applications due to its high freezing point (0 °C) and the risk of phase separation in microgravity, which can lead to inefficient heat transfer and system failure (Lahey and Dhir, 2004). Alternative working fluids, such as Na or K of SP-100, have been investigated for Rankine systems in space (Jiewei et al., 2024). The Stirling system has a lower mass compared to other

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Table 1
Parameter of some space nuclear reactors with power range of kW − MW.

Type	Nation	Year	Coolant	Dynamic system	Electric power/kW	Efficient/%
SNAP-SER (IECTE, 1984)	America	1959	NaK	TECS	5.0	10.0
HOMER-15 (Poston, 2001)	America	2001	Na	Stirling	3.0	20.0
AFSPS (Poston et al., 2008)	America	2006	NaK	Stirling	40.0	22.9
TEM (Hendrickx, 2019)	Russia	2009	He-Xe	Brayton	1000.0	28.6
MEGAHIT (Ruault et al., 2014)	EU	2013	He-Xe	Brayton	>1000.0	30.0
Prismatic space reactor (Yang et al., 2017)	China	2019	He-Xe	Brayton	1460.0	29.2

Table 2
Overview of System Analysis Codes for Space Reactor.

Code	Author	Year	Applications	Validation / Verification
DynMo (El-Genk and Tournier, 2005) DynMo-TE (El-Genk and Tournier, 2006) DynMo-CBC (El-Genk et al., 2009)	El-Genk et al.	2005 2006 2009	Space nuclear reactor power system	Validated against the design values from the Sectored Compact Reactors (SCoRes) (El-Genk and Tournier, 2006) and S^4-CBC space reactor (El-Genk et al., 2009)
SCTRAN (Wu et al., 2013)	XJTU-NUSOL	2013	Supercritical water or CO2 cooled reactor	Verified through RELAP5-3D and APROS for supercritical and subcritical condition (Wu et al., 2013) (Wu et al., 2012)
SIMCODE (Meng et al., 2021)	Tao Meng	2021	Gas-cooled space nuclear reactor system	–
SAC-SPACE (Zhang et al., 2021)	Zhang Ran	2021	Gas-cooled space nuclear reactor system	Validated against the design values from the Project Prometheus Space Nuclear Power Plant (SNPP). Verification includes comparisons with TRACE (Zhang et al., 2021)
SAMER (Qin et al., 2021)	Qin Hao	2021	Megawatt gas-cooled space reactor	Verified through the Computational Fluid Dynamics (Qin et al., 2021)

systems for low power demands (Mason and Carmichael, 2011). the Brayton system offers higher efficiency and a simple system structure for high power demands (Glassman, 1964). According to the literature, for megawatt-level demands, the gas-cooled Brayton system is more promising compared to other conversion system (Wu et al., 2025).

Ground gas-cooled Brayton systems typically use noble gas, specifically pure helium, as the working fluid. Compared to other noble gases, pure helium has high chemical stability, a small neutron absorption cross-section, greater specific heat capacity and thermal conductivity. However, its small molecular weight increases the aerodynamic loading and hence, the size and mass of the turbomachinery and heat exchanger.

Due to the requirements for mass and size in space reactors, pure helium is not suitable as the working fluid for space Brayton systems. El-Genk et al. (El-Genk and Tournier, 2007) investigated the advantages and limitations of using pure helium and noble gas binary mixtures as working fluids in closed Brayton systems. The study found that a He-Xe mixture gas with a molecular weight of 40 g/mol is the optimal working fluid. It has the same heat transfer performance as pure helium while reducing the consumed power on turbomachinery to only 10 % of that required for pure helium turbomachinery, which decrease the size and mass of the turbomachinery.

In recent years, many researches have been conducted in the development of system analysis code for He-Xe cooled space reactor. Table 2 shows the overview of system analysis codes for space reactor. El-Genk et al. (El-Genk and Tournier, 2005) developed a Dynamic simulation Model (DynMo) for space reactor power systems, which could be used for analyzing the transient and safety characteristics of the SRPSs with different types of nuclear reactors, energy conversion, and heat pipe radiators. Furthermore, safety analysis code DynMo-TE and DynMo-CBC are developed to simulate a space reactor power system with thermoelectric converters and multiple CBC Loops (El-Genk and Tournier, 2006) (El-Genk et al., 2009). Then, Gallo et al. (Gallo and El-Genk, 2009) designed and analyzed models of centrifugal-flow compressor and radial-inflow turbine of a 40.8 kWe Brayton cycle system using He-Xe working fluids with molecular weights of 15 g/mol and 40 g/mol. Zhou et al. (Biao et al., 2021) added a He-Xe property calculation module and flow heat transfer correlations to the system code RELAP5/MOD4.0. A one-dimensional space nuclear reactor system analysis code SIMCODE was developed (Meng et al., 2021), including turbine and compressor model, heat pipe radiator model. A system analysis for He-Xe cooled small reactor was developed (Wang et al., 2023) which improved the function of the original system analysis including property of He-Xe, the flow heat transfer model, and turbomachinery model. Zhang et al. (Zhang et al., 2021) developed the space nuclear system analysis code SAC-SPACE for the megawatt gas-cooled space reactor. Qin et al. (Qin et al., 2021) developed the space nuclear system analysis code (SAMER) by finite volume method and staggered grid technique for the preliminary safety analysis of the Open-grid megawatt gas cooled space nuclear reactor (OMEGA) conceptual design.

Many typical transient events for He-Xe cooled space reactor are calculated, including system startup, loss of heat sink accidents, loss of flow accident (LOFA), loss of coolant accident (LOCA), reactivity insertion accident (RIA) (Meng et al., 2021) (Wang et al., 2023) (Qin et al., 2021). El-Genk et al. (El-Genk and Tournier, 2006) (El-Genk et al., 2009) investigated the transient operation of the system during a startup and the load-following characteristic for S^4 and SCoRe-TE space reactor by using safety analysis code DynMo-TE and DynMo-CBC. Zhang et al. (Zhang et al., 2021) analyzed the safety characteristics of the system, with a focus on system response characteristics under the mechanical failure of one Brayton loop and reactivity insertion accidents. Wang et al. (Wang et al., 2021) utilized the system analysis code SAC-SPACE to investigate the safety characteristics of a gas-cooled space reactor Brayton cycle system under different operating conditions, which developed the conduction of the core and radiation heat transfer. Xi'an Jiaotong University Nuclear Safety and Operation Laboratory (XJTU-NUSOL) team also developed a dynamic analysis code SCTRAN designed for supercritical water or CO_2 cooled reactor. Dynamic models, including reactor, ducting, pump, recuperator, gas cooler and heat pipe radiator, are developed, which has been applied in studies the development of safety evaluation and control strategies for gas-cooled fast reactor and SCWRs (Gao et al., 2021) (Wang et al., 2024).

Based on the above researches, it can be concluded that space reactor systems are usually arranged with two or more loops to avoid single-point failures. In studies of the safety characteristics of space reactor

systems during accidents, many studies have primarily focused on scenarios such as loss of heat sink and loss of flow in the overall system, with limited attention given to the safety characteristics under accidents involving one or two loops. Therefore, in this paper, based on the present system analysis code SCTRAN, the physical property, flow and heat transfer correlations, as well as the turbomachinery model for He-Xe are developed and validated in Section 2. Next, a steady-state model for S^4 (Submersion, Subcritical Safe Space) reactor of the three-loop system is established in Section 3. The S^4 space reactor was selected for this study due to its unique combination of high efficiency, compact design, and robust safety features, making it particularly suitable for long-duration, high-power space missions (King and El-Genk, 2006). Unlike liquid–metal-cooled reactors, which pose risks of liquid droplet contamination in orbital environments, the S^4 employs a gas-cooled reactor with a He-Xe working fluid, eliminating such concerns. Its modular design, featuring three independent cooling loops, ensures redundancy and mitigates single-point failures, enhancing system reliability. Additionally, its passive safety mechanisms, including negative temperature reactivity feedback and subcritical safety in submerged conditions, further distinguish it from alternative designs (King and El-Genk, 2006). Finally, the safety characteristics of the S^4 reactor under typical accidents are investigated in Section 4.

2. Model development and validation

2.1. Introduction to SCTRAN

SCTRAN is a 1D safety analysis code, containing multi-group point-kinetics equations and other component models for supercritical water or CO_2 cooled reactors (Wu et al., 2020). It also has heat transfer models and heat conduction that allow the modelling of the components. SCTRAN has been used to conduct design studies for SCWRs and can simulate dynamic events involving supercritical and subcritical states (Wu et al., 2014). The trans-critical capability of SCTRAN has been validated by comparing its results with those of the RELAP5-3D code and experimental data (Wu et al., 2012). It has been applied to Supercritical CO_2 (SCO_2) Brayton cycle coupled nuclear system in many studies regarding transient and safety analysis (Gao et al., 2021). In this work, the physical property of He-Xe, flow resistance and heat transfer correlation, as well as performance curve for He-Xe applied turbomachinery models are needed to be developed in this paper.

2.2. Differential equations

RETRAN − 02 is a one − dimensional safety analysis code approved by the United States Nuclear Regulatory Commission for licensing calculations (McFadden et al., 1988). The SCTRAN code references the numerical discretization methods of RETRAN − 02, and in addition to that, incorporates more accurate auxiliary and constitutive equations, including equations for water and CO_2 properties, heat transfer correlations, resistance correlations, critical heat flux models, critical flow models, and wall friction coefficient model (Wu et al., 2013) (Wu et al., 2018). The basic equations of SCTRAN include mass conservation equation, momentum conservation equation, and energy conservation equation.

The mass equation is shown in formula (1).

$$\frac{\partial}{\partial t}\rho A+\frac{\partial}{\partial z}W=0 \tag{1}$$

Where, A is the flow area, m^2. ρ is mixture density of vapor and liquid phases calculated by formula (2), kg/m^3. W is mixture mass flow rate of vapor and liquid phases calculated by formula (3), kg/s.

$$\rho=\alpha_g\rho_g+\alpha_l\rho_l \tag{2}$$

$$W=W_g+W_l=(\alpha_g\rho_g V_g+\alpha_l\rho_l V_l)A \tag{3}$$

Where, V is average fluid velocity, m/s. The subscript g is vapor phase and the subscript l is liquid phase.

The momentum equation is shown in formula (4).

$$\frac{\partial}{\partial t}W+\frac{\partial}{\partial z}WV=-A\frac{\partial P}{\partial z}-\frac{2A\rho V|V|}{D_h}f_{tp}+\rho A g_z \tag{4}$$

Where, the first term on the right side is the pressure difference where P is pressure, Pa. the second term is the Fanning friction pressure drop where f_{tp} is Fanning friction factor, D_h is hydraulic diameter, m. the third term is the gravity pressure drop where g_z is axial component of acceleration due to gravity, m/s^2.

The energy equation is shown in formula (5).

$$\frac{d}{dt}U=-\frac{1}{2}\frac{L}{A}\frac{d}{dt}\left(\frac{W^2}{\rho}\right)-\sum_j\left\{(W_g h_g+W_l h_l)+\frac{1}{2}(W_g V_g V_g+W_l V_l V_l)+Wg(z-z_j)\right\}+Q \tag{5}$$

The energy equation is a compressible formulation. Where, $U=\int_V\rho(e+\widehat{Z})dV$ is internal energy per unit volume, e is energy per unit mass, $\widehat{Z}$ is gravitational potential, J/kg. h_g and h_l are specific enthalpy of vapor and liquid phase in the control volume, J/kg. z is vertical distance above reference base and z_j is junction height above reference height z, m. Q is the energy change caused by heat transfer and internal heat source, W/m^3.

2.3. Difference equations

A numerical scheme having a staggered spatial mesh is applied in SCTRAN, which the velocities at boundaries are most conveniently defined through use of momentum control volumes centered(cell) on the mass and energy cell boundaries. The energy conservation equation and mass conservation equation are applied to the control volumes to calculate the changes in mass, internal energy, and pressure. The momentum conservation equation is applied to the junction to calculate the mass flow rate inlet and outlet of the control volumes. The finite difference equations for the mass, momentum and energy balances are listed below.

The mass equation for control volume K is shown in formula (6).

$$\frac{dM_K^{n+1}}{dt}=W_{j-1}^{n+1}-W_j^{n+1} \tag{6}$$

Where, $M_k=\rho A$, the superscripts n, $n+1$ are time level index. The two terms on the right side are the mass flow rate inlet and outlet of the control volume K.

The momentum equation for junction j is shown in formula (7).

$$\left(\frac{l_k}{2A_K}+\frac{l_L}{2A_L}\right)\frac{dW_j^{n+1}}{dt}=\left(\frac{1}{\rho_K A_K^2}\right)^n\left(W_j^2\right)^{n+1}-\left(\frac{1}{\rho_L A_L^2}\right)^n\left(W_j^2\right)^{n+1}+\left\{\frac{1}{2\rho_j}\left(\frac{1}{A_L^2}-\frac{1}{A_K^2}\right)\right\}^n\left(W_j^2\right)^{n+1}+(P_K-P_L)^{n+1}-F_j^n|W_j|^n W_j^{n+1}-\left(\frac{K_j}{2\rho_j A_j^2}\right)^n|W_j|^n W_j^{n+1}-\left(\int_K^j\rho dz+\int_j^L\rho dz\right)^n g \tag{7}$$

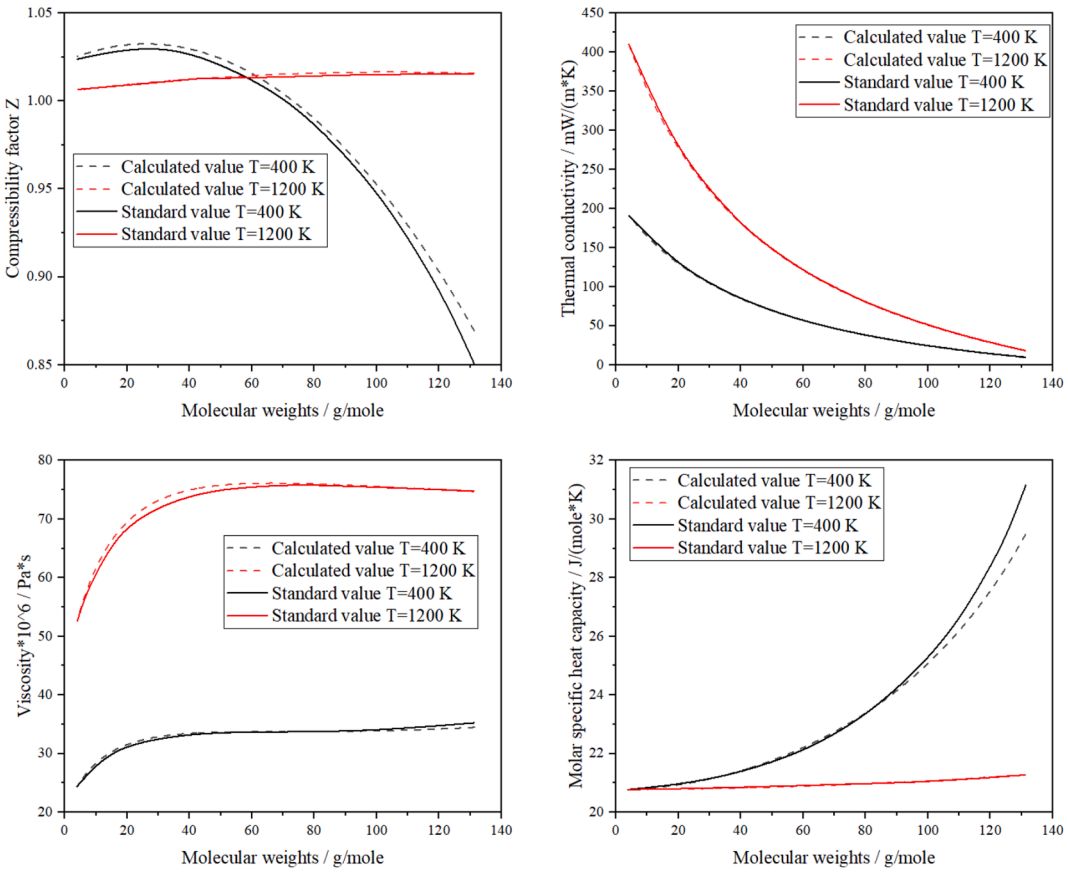


Fig. 1. Comparison of calculated values and standard values from Tournier (El-Genk and Tournier, 2008) of four physical properties under different molecular weights.

Where, the subscript *K*, *L* are control volume *K* and *L*. the subscript *j* is junction *j*. *l* is length of control volume. the first and second terms on the right side are the momentum changes caused by liquid compression and changes in the control volume area, the third term is the momentum changes caused by changes in the junction area, the fourth term is the pressure difference between adjacent control volume, the fifth term is the friction pressure drop of the half control volume adjacent to the junction, the sixth term is the friction pressure drop of the junction, and the seventh term is the elevation pressure drop of the control volume adjacent to the junction.

The energy equation for control volume *K* is shown in formula (8).

$$\frac{dU_K^{n+1}}{dt} = (W_{j-1})^{n+1}\left\{h_{j-1}^n + \frac{1}{2}\left[\left(\frac{W_{j-1}^2}{\rho_{j-1}A_{j-1}}\right)^2\right]^n + g(z_{j-1} - z_K)\right\} - (W_j)^{n+1}\left\{h_j^n + \frac{1}{2}\left[\left(\frac{W_j^2}{\rho_j A_j}\right)^2\right]^n + g(z_j - z_i)\right\} - \frac{l_K}{2A_K}[\frac{d}{dt}(\frac{(\frac{1}{2}(W_{j-1}+W_j))^2}{\rho_K})]^n + Q_K^n \quad (8)$$

Where, the first and second terms on the right are the energy inlet and outlet the control volume, the third term is the rate of change in kinetic energy of the control volume, where the mass flow rate of the control volume is equal to the average flow rate of all connected junctions, and the fourth term is the internal heat source of the control volume.

2.4. Thermo-physical properties for He-Xe

In this section, based on the semi-empirical formulas of physical property for noble gas mixture proposed by Tournier (Tournier et al., 2006); a further development of the property calculation code for the He-Xe mixture gas is undertaken on the present system analysis code SCTRAN. The partial semi-empirical formula of physical property for noble gas mixture are provided as following.

The calculated temperature range of He-Xe mixture gas is 300 K-1200 K, and the pressure range is 0.1Mpa-20Mpa. For the actual gas equation of state:

$$Z = \frac{p\nu}{R_g T} = 1 + \frac{B}{\nu} + \frac{C}{\nu^2} + \frac{D}{\nu^3} + \ldots \quad (9)$$

Where, ν is the specific volume of gas, m^3/kg. *B* denotes the second virial coefficients, $mole/m^3$. *C* denotes the third virial coefficients, $mole^2/m^6$. *D* denotes the fourth virial coefficients, $mole^3/m^9$. In this paper, the second virial coefficients, third virial coefficients are used to

Table 3
Four heat transfer correlations.

Year	Author	Formula	*Re*	*Pr*	D_h
1930	D-B (Dittus, 1930)	$Nu = 0.023Re^{0.8}Pr^{0.4}$	$10^4 \sim 1.2 \times 10^5$	$0.7 \sim 120$	–
1968	Taylor (Taylor, 1968)	$Nu_b = 0.023Re_b^{0.8}Pr^{0.65}(T_w/T_b)^{-n}$ $n = (0.57 - 1.59/(x/D))$	$1.8 \times 10^4 \sim 6 \times 10^4$ $T_w/T_b < 2$	–	–
1970	Petukhov (Petukhov, 1970)	$Nu = \frac{(\xi/8)RePr}{K_1(\xi) + K_2(Pr)\sqrt{(\xi/8)}(Pr^{2/3} - 1)}$ $\xi = (1.82 \cdot \log Re - 1.64)^{-2}$ $K_1(\xi) = 1 + 3.4\xi$ $K_2(Pr) = 11.7 + 1.8Pr^{-1/3}$	$10^4 \sim 5 \times 10^6$	$0.5 \sim 200$	–
2022	Wang (Qin et al., 2022)	$Nu = 0.016Re^{0.85}Pr^{0.77}$	$\geq 10^4$	–	9 mm

analyze the He-Xe mixture gas.

The density of the He-Xe mixture gas is calculated by formula (10).

$$\rho = \frac{p}{ZR_gT} \tag{10}$$

Where, $R_g = R/M$ is the gas constant. $R = 8.31441$, J/(mole·K). $M = xM_1+(1\text{-}x)M_2$ is the average molecular weights of the He-Xe mixture gas, kg/mole.

The molar specific heat capacity at constant pressure is calculated by formula (11).

$$\widehat{C}_p = \widehat{C}_{p_0} + \rho R_g\left[\left(B - T\frac{dB}{dT} - T^2\frac{d^2B}{dT^2}\right) + \overline{\rho}\left(C - \frac{T^2}{2}\frac{d^2C}{dT^2}\right)\right] - R_gT[(B - T\frac{dB}{dT}) + \overline{\rho}(2C - T\frac{dC}{dT})]\frac{(\overline{\rho} + B\overline{\rho}^2 + C\overline{\rho}^3)/T + \frac{dB}{dT}\overline{\rho}^2 + \frac{dC}{dT}\overline{\rho}^3}{1 + 2B\overline{\rho} + 3C\overline{\rho}} \tag{11}$$

Where, $\widehat{C}_p$ is specific heat capacity at constant pressure, J/(mole·K). $\widehat{C}_{p_0} = 5R/2$ is the ideal values for a dilute gas. $\overline{\rho} = \rho/M$ is molar density, mole/m^3.

The molar specific heat capacity at constant volume is calculated by formula (12).

$$\widehat{C}_v = \widehat{C}_{v0} - \overline{\rho}R_gT\left[\left(2\frac{dB}{dT} + T\frac{d^2B}{dT^2}\right) + \overline{\rho}\left(\frac{dC}{dT} + \frac{T}{2}\frac{d^2C}{dT^2}\right)\right] \tag{12}$$

Where, $\widehat{C}_v$ is specific heat capacity at constant volume, J/(mole·K). $\widehat{C}_{v0} = 3R/2$ is the ideal values for a dilute gas.

The viscosity is calculated by formula (13).

$$\mu_{\text{mix}} = \mu^o + \left(1 - \frac{1}{2.3}\right)\mu^* \times \Psi_\mu\left(\frac{0.291 \times V^*}{M}\rho\right) \tag{13}$$

Where u^0 is the weighted average of viscosity, Pa·s. And the latter is corrected for ideal gas. Ψ_u is the test data fitting function, whose expression is as follows:

$$\Psi_\mu(\rho_r) = 0.221\rho_r + 1.062\rho_r^2 - 0.509\rho_r^3 + 0.225\rho_r^4 \tag{14}$$

Where, $\rho_r = \rho/\rho_{cr}$ is reduced density. ρ_{cr} is critical density.

u^* is the fitting viscosity calculated by formula (15).

$$\mu^* = 0.204 \times 10^{-7}\frac{\sqrt{MT^*}}{(0.291 \times V^*)^{\frac{2}{3}}} \tag{15}$$

Where, T^* and V^* are the pseudo-critical temperature and molar volume of the binary gas mixture.

The thermal conductivity is calculated by formula (16).

$$\lambda_{\text{mix}} = \lambda^o + \left(1 - \frac{1}{2.94}\right)\lambda^* \times \Psi_\lambda\left(\frac{0.291 \times V^*}{M}\rho\right) \tag{16}$$

Where λ^0 is the weighted average of thermal conductivity, W/(m·K). And the latter is corrected for ideal gas. Ψ_λ is the test data fitting function, whose expression is as follows:

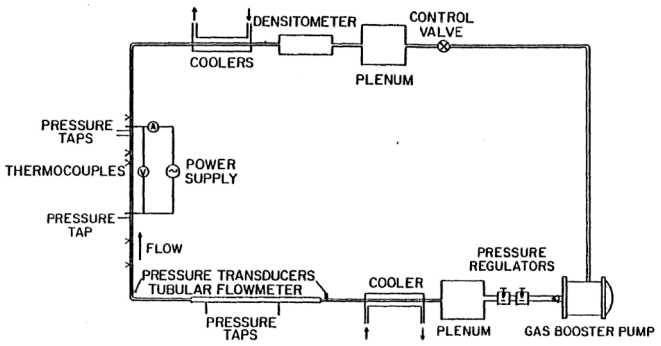


Fig. 2. Schematic diagram of circular tube heating experimental device (Taylor et al., 1988).

$$\Psi_\lambda(\rho_r) = 0.645\rho_r + 0.33\rho_r^2 + 0.0368\rho_r^3 - 0.0128\rho_r^4 \tag{17}$$

λ^* is the fitting thermal conductivity calculated by formula (18).

$$\lambda^* = 0.304 \times \frac{10^{-4}(T^*)^{0.227}}{M^{0.465}(0.291V^*)^{0.415}} \tag{18}$$

Basing on Fortran 95 language, the calculation code for thermo-physical properties of He-Xe is developed. The compressibility factor, thermal conductivity, viscosity and molar heat capacity of different molecular weights at 2 MPa, 400 K and 1200 K were calculated and compared with the standard values from the reference (El-Genk and Tournier, 2008).

Fig. 1 shows a comparison of calculated values and standard values of four physical properties under different molecular weights. The compressibility factor slightly increases first and then decreases at 400 K while it slightly increases at 1200 K with the increase of molecular weights. The thermal conductivity decreases with the increase of molecular weights at 400 K and 1200 K. The viscosity increases first and then slightly decreases at 1200 K while it increases at 400 K with the increase of molecular weights. Molar heat capacity increases with the increase of molecular weights at 400 K while remain almost constant at 1200 K. It can be seen that the calculated values from the developed physical property calculation code closely align with the standard values, thereby validating the accuracy of development of the physical property for He-Xe.

2.5. Selection of flow resistance and heat transfer correlation based on experimental data

Due to the fact that the object in this paper is a circular channel for coolant, a suitable flow and heat transfer correlation for He-Xe circular tube needs to be selected. In the section, selecting the circular tube heating experiment by Taylor (Taylor et al., 1988) for modeling, four heat transfer correlations shown in Table 3 were evaluated. Fig. 2 shows the device of circular tube heating experiment, in which the

Table 4
Experimental parameters of He-Xe mixture gas (Taylor et al., 1988).

Serial number	Molecular weights /g/mol	Pr	Re	q /$W{\cdot}m^{-2}$	G /$kg{\cdot}m^{-2}{\cdot}s^{-1}$	Outlet pressure/Pa
Run687	83.8	0.25	63987 ~ 85802	43,637	350.6	471,502
Run696	39.5	0.21	19174 ~ 36183	136,770	156.4	563,474
Run707	28.3	0.23	34443 ~ 53390	157,094	229.1	928,019
Run715	14.5	0.30	19485 ~ 34042	296,622	139.7	806,581

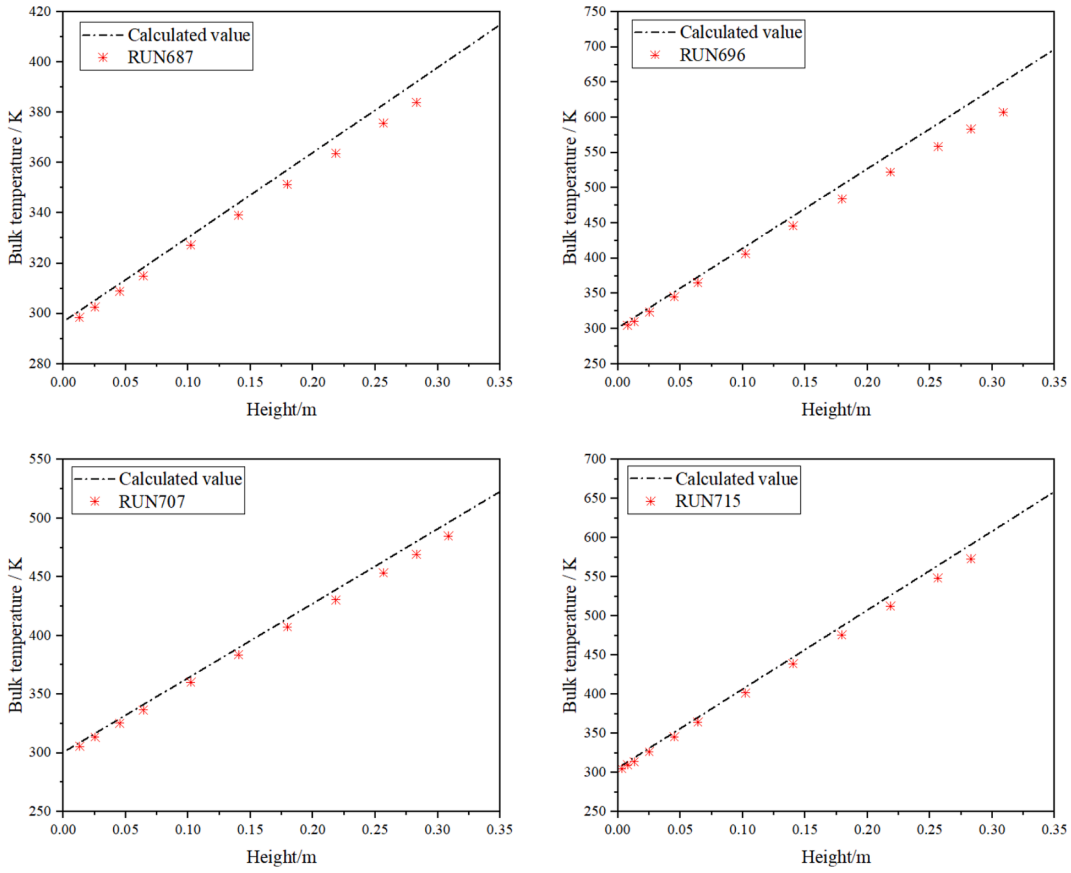


Fig. 3. Comparison between calculated bulk temperature and experimental data.

experimental section is a circular tube with an inner diameter of De = 5.87 mm. It consists of an insulation section with length of 56 De and a uniform heat flux section length of 60 De. The experiment explores the flow heat transfer process of various noble gas mixtures, including He-Xe mixture gas.

A total of 53 runs were performed cover various noble gas mixtures such as He-Xe, H_2-Xe and air. As shown in Table 4, To cover the He-Xe molecular weight range currently used in space reactors, four experiments for flow heat transfer calculation with different molecular weights are selected. The results are shown in Figs. 3–5. Under the same experiment, different heat transfer correlations only change the wall temperature rather than the bulk temperature. Fig. 3 shows the calculated values and experimental values of bulk temperature under various operating conditions. It can be seen that there are some differences between the calculated values and experimental values. Because the He-Xe mixture gas of Taylor's experiment (Taylor et al., 1988) was assumed to be ideal gases while the properties in this paper are calculated by the semi-empirical formulas proposed by Tournier (Tournier et al., 2006).

Fig. 4 shows the comparison between calculated wall temperature and experimental data. It can be found that the D-B correlation predicts the lowest wall temperature, because the applicable range of the Prandtl number is between 0.7 and 120. However, He-Xe is low Prandtl fluid ranging between 0.2 and 0.3 (As shown in Table 4). At the beginning of the experimental heating stage, the thermal boundary layer is thin and heat transfer is intense. Wall temperature predicted by the Taylor correlation is closest to the experimental data. According to Table 3, the Taylor correlation has been modified for the inlet section, making it closer to the experimental data. During the thermally and hydrodynamically fully developed stage, the coolant reaches a thermally stable state. Under these conditions, the wall temperature exhibits a consistent and predictable linear increase over length due to the steady heat flux applied at the boundary. The Wang and Petukhov correlation also predicts a relatively close wall temperature.

Fig. 5 shows the comparison between calculated heat transfer coefficient and experimental data. The heat transfer coefficient predicted by the D-B correlation is too high, with most errors exceeding 50 %. When the heat transfer coefficient is low ($\leq 500 W{\cdot}m^{-2}{\cdot}K^{-1}$), the Taylor, Wang, and Petukhov correlation perform well with errors generally within 20 %; When the heat transfer coefficient is relatively high ($\geq 500 W{\cdot}m^{-2}{\cdot}K^{-1}$), the Taylor correlation predicts the best, with errors within 20 %.

The pressure drop in SCTRAN is calculated by Blasius correlation, which is shown by formula (19):

$$f = 0.3164/Re^{0.25} \tag{19}$$

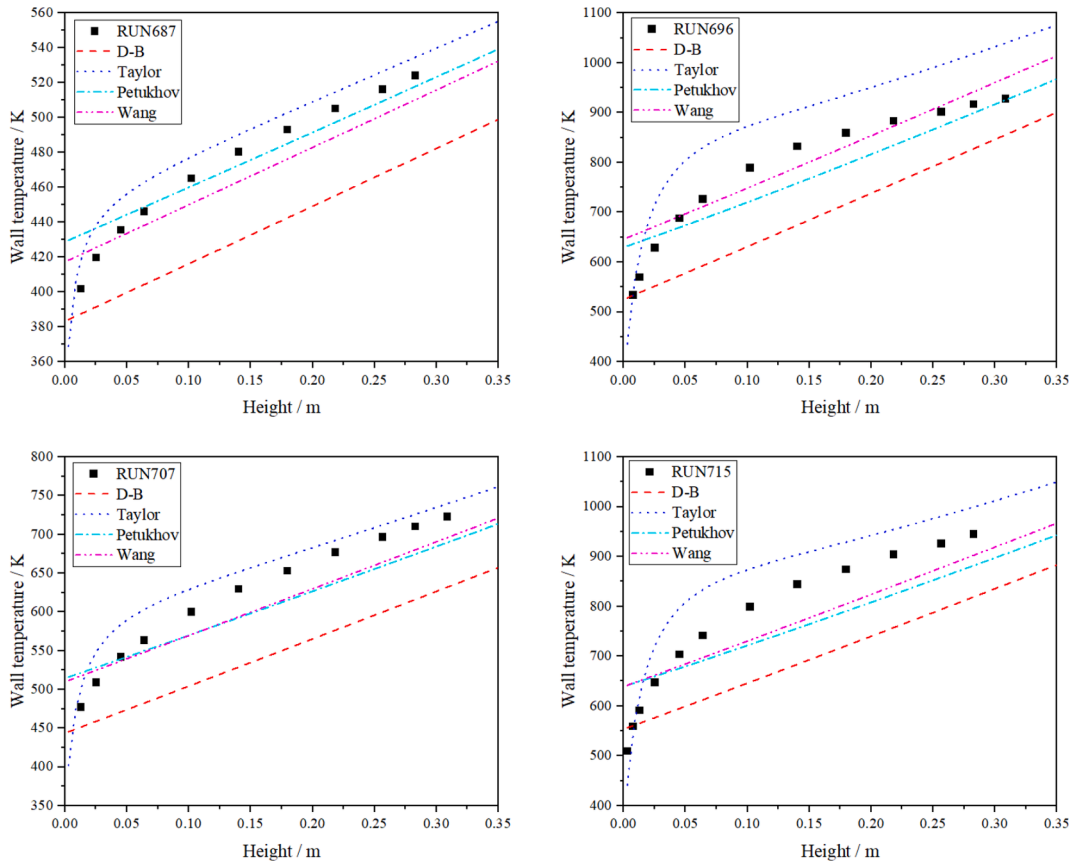


Fig. 4. Comparison between calculated wall temperature and experimental data.

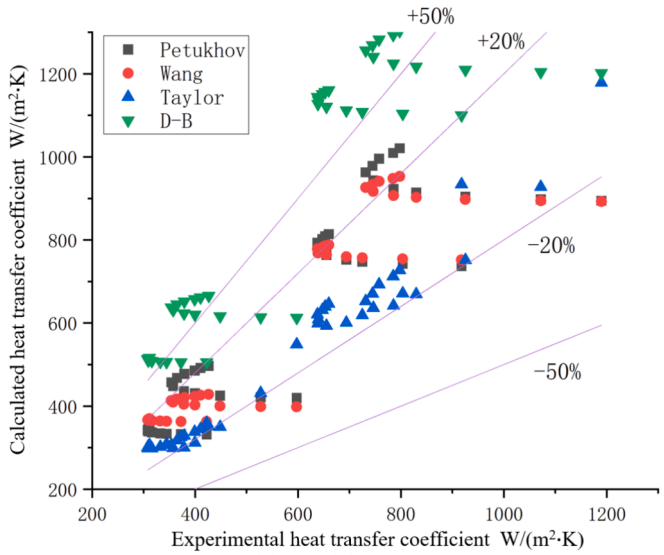


Fig. 5. Comparison between calculated heat transfer coefficient and experimental data.

Table 5 shows the error of calculated pressure drop for He-Xe mixture gas. It can be seen that the error of pressure drop calculation for four operating conditions is within 10 %, which validates that the Blasius correlation can predict the pressure drop of He-Xe.

Through the above calculations, the D-B correlation is no longer applicable for the He-Xe, while the Taylor correlation is the most consistent with experimental data. Therefore, the Blasius and Taylor correlation will be used for flow and heat transfer calculations in the future research.

Table 5
Error of calculated pressure drop for He-Xe.

Serial number	Experimental value (Taylor et al., 1988) / Pa	Calculated value / Pa	Error
Run687	7584	7325	3.4 %
Run696	5515	5415	1.8 %
Run707	6894	6504	5.7 %
Run715	8273	7951	3.9 %

2.6. Turbomachinery models development and validation

2.6.1. Compressor model

The solution of compressor model should include pressure rise which could be used for fluid momentum conservation equation, enthalpy increase which was needed in fluid energy conservation equation, and torque which is needed for shaft model to simulate rotating speed.

Fig. 6 shows the fluid enthalpy and entropy variation during ideal and realistic compression process. The adiabatic efficiency of the compressor is defined as:

$$\eta_{ad} = \frac{\text{Isentropic work}}{\text{Actual work}} = \frac{h^T_{2,\text{ideal}} - h^T_1}{h^T_{2,\text{real}} - h^T_1} \tag{20}$$

Where, $h^T_{2,\text{real}}$ is the real enthalpy at the compressor outlet, $h^T_{2,\text{ideal}}$ is the ideal enthalpy at the compressor outlet, h^T_1 is the real enthalpy at the compressor inlet.

Therefore, the fluid enthalpy increases:

$$\Delta h = \frac{h^T_{2,\text{ideal}} - h^T_1}{\eta_{ad}} \tag{21}$$

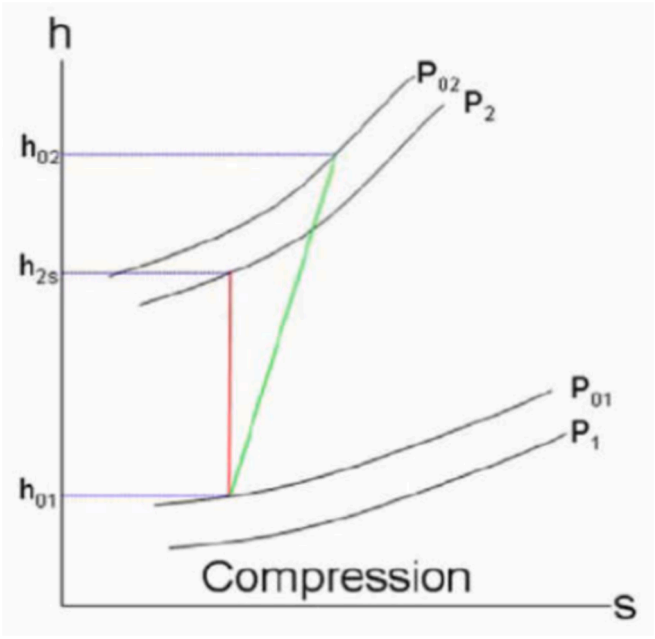


Fig. 6. Ideal and realistic compression process inside compressor.

The compressor head applied to the momentum equation is:

$$\Delta P = \rho \cdot g \cdot H = P_1^T\left(R_p - 1\right) = P_2^T\frac{R_p - 1}{R_p} \tag{22}$$

Where, R_p is the compressor pressure ratio, P_1^T is the compressor inlet total pressure, P_2^T is the compressor outlet total pressure.

The total torque of the compressor is obtained:

$$\tau_t = \tau_s + \tau_d = \frac{\dot{m}}{\omega}\frac{1}{\eta_{ad}}\left(h_{2,\text{ideal}}^T - h_1^T\right) \tag{23}$$

Where, τ_s is compressor torque during isentropic process, τ_d is dissipated torque, N·m. $\dot{m}$ is mass flow rate of compressor working fluid. ω is shaft speed, rad/s.

2.6.2. Shaft model

The angular momentum equation of the shaft based on the torque of turbomachinery on the rotating shaft is calculated. The shaft model is shown in formula (24):

$$\sum_i I_i\frac{d\omega}{dt} = \sum_m \tau_{T,m} - \sum_n \tau_{C,n} + \tau_g \tag{24}$$

Among them, I_i is moment of inertia, kg·m^2.The first term on the right side of the formula (24) denotes all turbine torques on the shaft, the second term represents all compressor torques connected to the shaft, and the third term represents the torque of the alternator.

Table 6
Comparison between calculated and design values.

Parameter	Design value (Gallo and El-Genk, 2009)	Calculated value	Error
Turbine inlet temperature/K	1149	1148.5	0.04 %
Turbine outlet temperature /K	–	1011.9	–
Compressor inlet temperature/K	400	400.1	0.03 %
Compressor outlet temperature /K	483.3	482.5	0.17 %
Turbine inlet pressure/ kPa	902.4	888.6	1.53 %
Turbine outlet pressure/ kPa	635.5	623.9	1.83 %
Compressor inlet pressure/kPa	621.1	620.9	0.03 %
Compressor outlet pressure/kPa	931.6	931.4	0.02 %

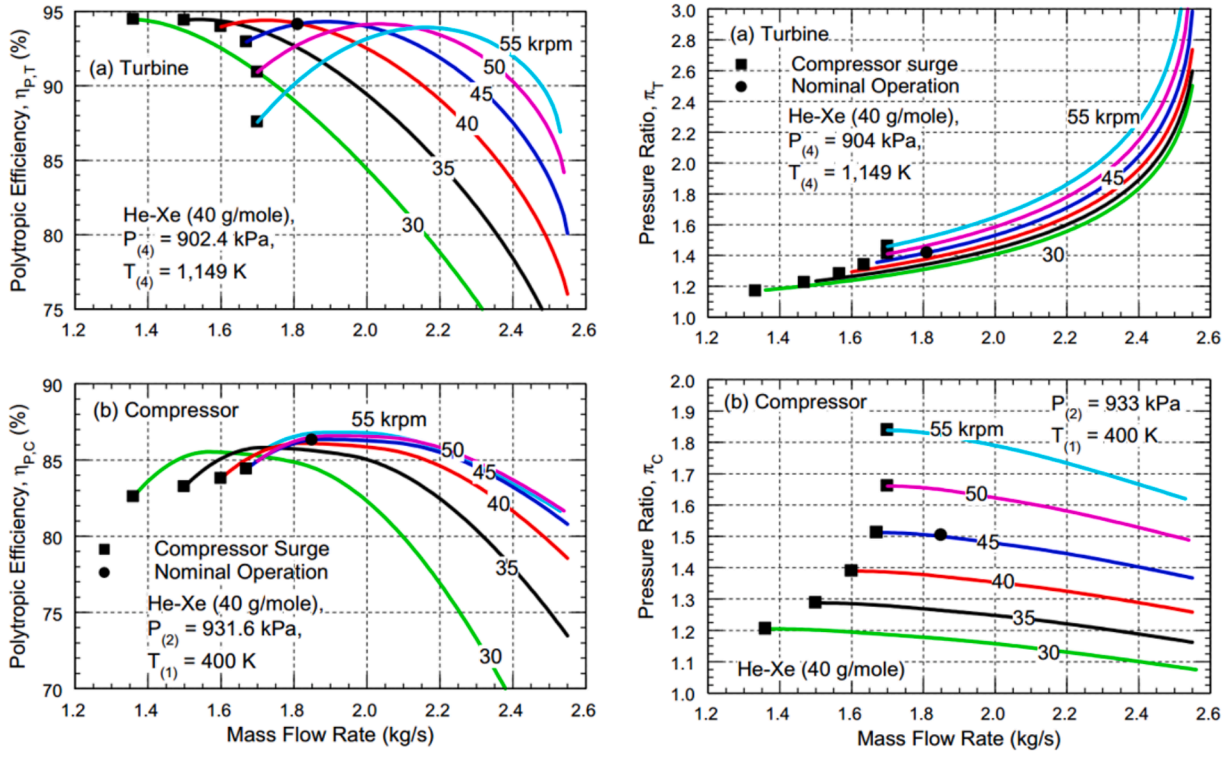


Fig. 7. Performance Curve of 40 g/mol He-Xe turbomachinery (Gallo and El-Genk, 2009).

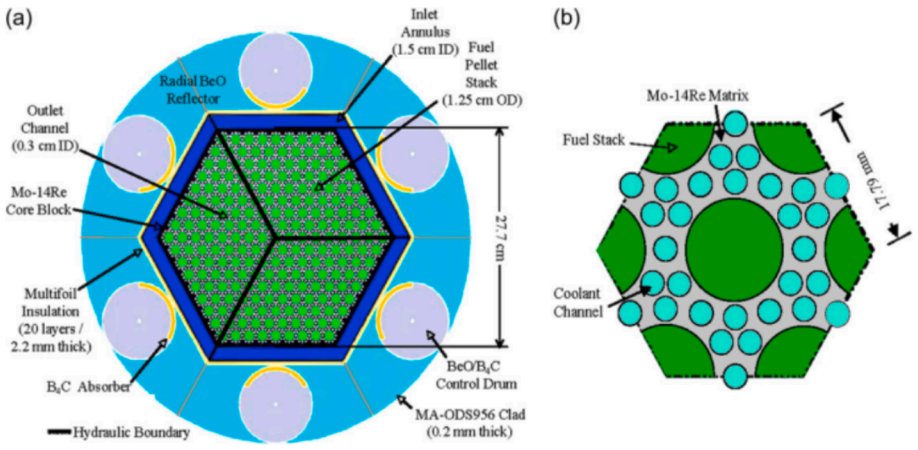


Fig. 8. Radial cross-sections of (a) S^4 reactor core and (b) Unit cell (King and El-Genk, 2009).

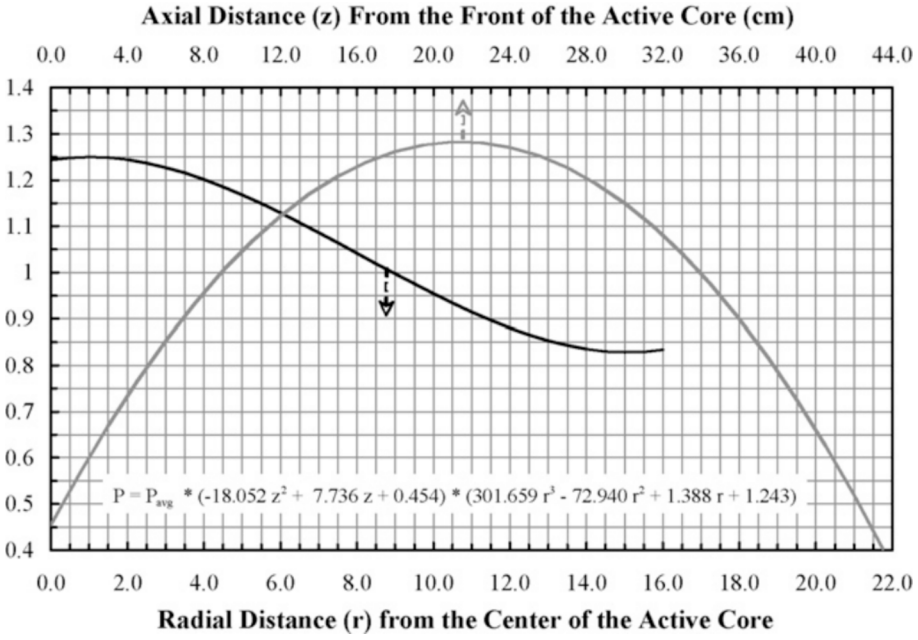


Fig. 9. Axial and radial fission power distributions in S^4 reactor core (King and El-Genk, 2009).

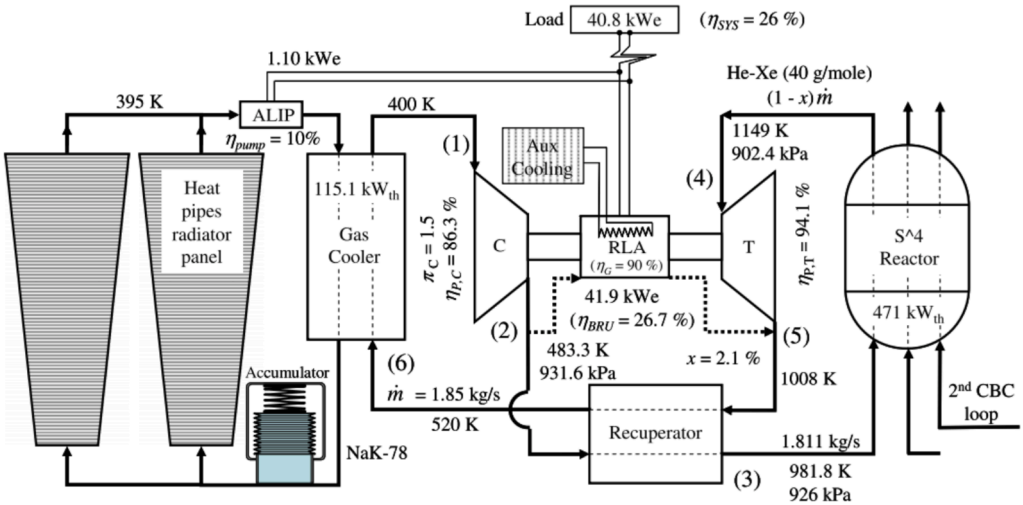


Fig. 10. Steady-state operation parameters of a cbc loop in s^4 reactor (Gallo and El-Genk, 2009).

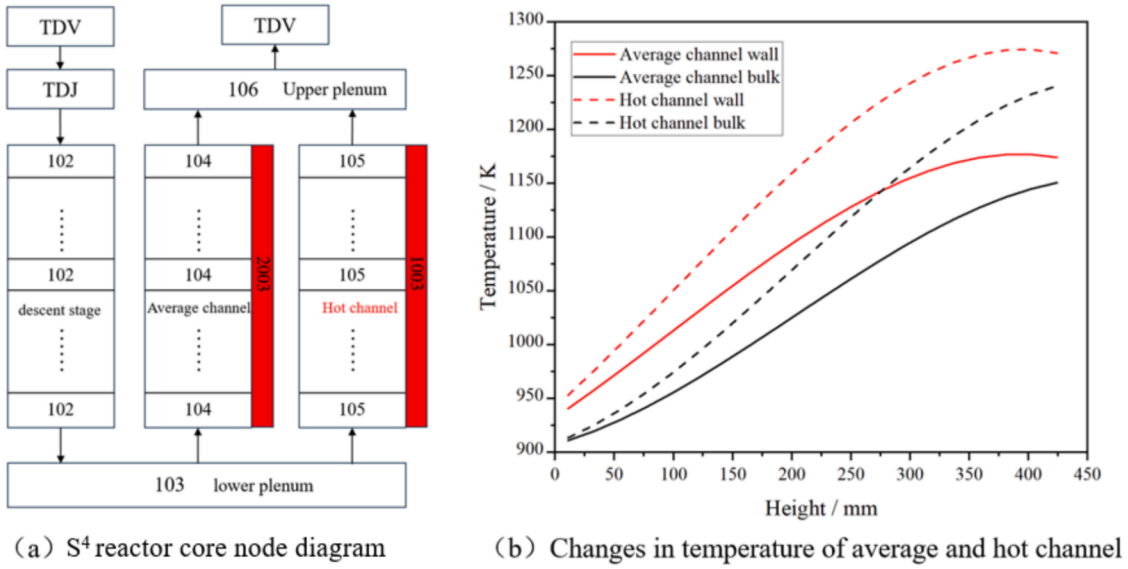


Fig. 11. S^4 reactor core node diagram and temperature distribution.

Table 7
Comparison between calculated and design values of the S^4 reactor core.

Parameter	Design value (King and El-Genk, 2009)	Calculated value	Error
Core inlet temperature/K	904	904.6	0.06 %
Core outlet temperature/K	1149	1151.6	0.07 %
Total mass flow rate/$kg{\cdot}s^{-1}$	3.67	3.67	0.00 %
Maximum coolant temperature/K	1284	1240	3.43 %
Maximum fuel temperature/K	1302	1282.7	1.48 %
Core inlet pressure/MPa	1.985	1.934	0.25 %

2.6.3. *Performance curve*

As mentioned above, El-Genk et al. (Gallo and El-Genk, 2009) (El-Genk et al., 2006) have conducted extensive research on the turbomachinery performance curve for He-Xe. Fig. 7 shows the relationship between the pressure ratio and efficiency of the 40 g/mol He-Xe working fluid with mass flow rate and speed. The efficiency of the turbine at a rated mass flow rate of 1.811 kg/s and a rated speed of 45 krpm is 94.1 %; with the pressure ratio of 1.42; The efficiency of the compressor at a rated mass flow rate of 1.85 kg/s and a rated speed of 45 krpm is 86.4 %, with the pressure ratio of 1.5. In this paper, the performance curve for He-Xe is added to the SCTRAN to calculate the parameters of compressor and turbine.

2.6.4. *Model validation*

According to the rated parameters mentioned above, steady-state inlet and outlet boundary conditions are set for the turbine and

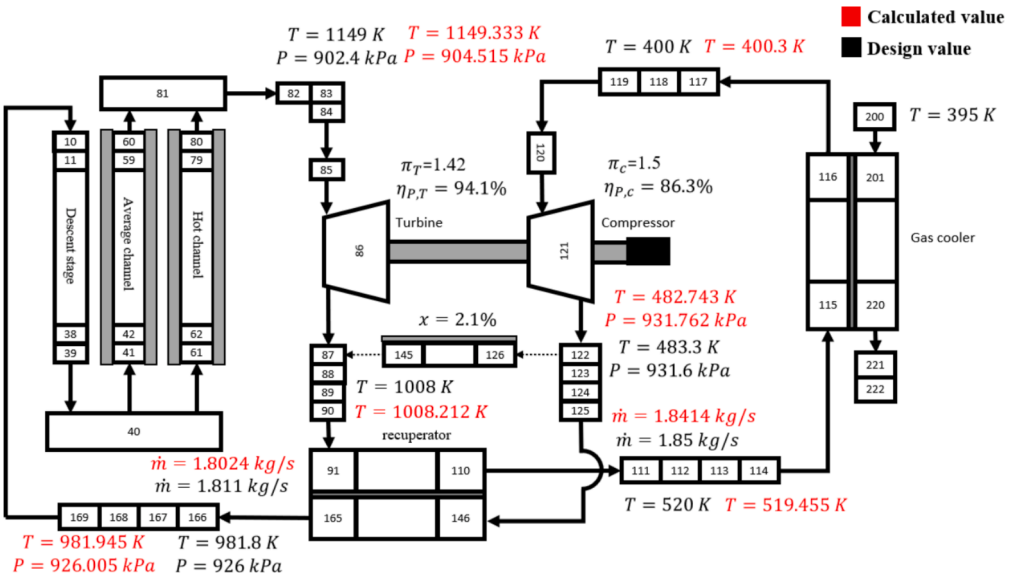


Fig. 12. Node diagram of He-Xe Brayton cycle cooled S^4 reactor system.

Table 8
Comparison between calculated and design values of system steady-state.

Parameter	Design value (Gallo and El-Genk, 2009)	Calculated value	Error
Core inlet mass flow rate/kg/s	1.811	1.8024	0.47 %
Core inlet temperature/K	981.8	981.945	0.01 %
Core inlet pressure/kPa	926	926.005	0.00 %
Core outlet temperature/K	1149	1149.333	0.03 %
Core outlet pressure/kPa	902.4	904.515	0.23 %
Inlet temperature of hot section of regenerator/K	1008	1008.212	0.02 %
Outlet temperature of hot section of regenerator /K	520	519.455	0.10 %
Mass flow rate of hot section of regenerator / kg/s	1.85	1.8414	0.46 %
Compressor inlet temperature/ K	400	400.3	0.08 %
Compressor outlet temperature/K	483.3	482.743	0.12 %
Compressor outlet pressure/ kPa	931.6	931.762	0.02 %

compressor. The specific boundary conditions are as follows: for the turbine the inlet temperature is 1149 K, inlet pressure is 902.4 kPa, outlet temperature is 1012 K, outlet pressure is 622 kPa, pressure ratio is 1.42, and efficiency is 94.1 %; for the compressor, the inlet temperature of is 400 K, the inlet pressure is 621 kPa, the outlet temperature is 483 K, the outlet pressure is 931.6 kPa, the pressure ratio is 1.5, and the efficiency is 86.4 %. As shown in Table 6, the maximum error of calculation result is within 2 %, which verifies the accuracy of the development of the turbomachinery models.

3. Numerical modelling of S^4 reactor

3.1. S^4 reactor description

The S^4 (Submersion, Subcritical Safe Space) reactor power system is designed to operate for 12 years, whose thermal power is 471 kW. The working fluid of the S^4 reactor is He-Xe with 40 g/mol, which serves not only as the coolant for the reactor but is also used for energy conversion (King and El-Genk, 2009). Radial schematic diagram of the S^4 reactor core is shown as Fig. 8(a), which is divided into three sectors to avoid single-point failures. And each sector is coupled to a closed Brayton loop, which is hydraulically independent but thermally and neutronically coupled. Fig. 8(b) shows the design of the unit cells in S^4 reactor core. Each fuel pin with a diameter of 1.25 cm is surrounded by 12 coolant channels with a diameter of 3 mm, and the fuel pins are arranged in a triangular lattice with a pitch of 1.779 cm. Fig. 9 shows axial and radial fission power distributions in the S^4 reactor core. Fig. 10 shows the diagram of one of the three CBC loops of the S^4 reactor during steady-state operation. It can be seen that the CBC loops have the separate heat pipes radiator panel. The efficiency of each loop is 26 % with the output power of 40.8 kW.

3.2. Reactor model and validation

Based on the parameters above, the model of the S^4 reactor core is analyzed by using the developed system analysis code. Fig. 11 shows the node diagram and temperature distribution of the S^4 reactor core. According to the axial and radial power distribution within the core, the coolant channels are divided into average channel and hot channel. The wall temperature first increases and then decreases, while the bulk temperature increases. The maximum wall temperature (1274.6 K) occurs at the eighteenth control volume in the axial height, which is consistent with the reference (King and El-Genk, 2009). Comparison between calculated and design values of the S^4 reactor core is shown in Table 7. It can be observed that the maximum coolant temperature is 1240 K and the maximum fuel temperature is 1282.7 K. With the maximum error within 4 %; thereby the accuracy of the core model is validated.

Table 9
Safety design limits for different typical space reactor schemes.

Core type	SP-100 (Truscello and Rutger, 1992)	SPACE-R (Rhee et al., 1992)	HOMER (Poston, 2001)
fuel	UN	UO_2	UN
cladding material	Re/Nb-1Zr	W	Stainless steel
melting point of fuel/*K*	3123	3120	3123
Melting point of cladding /*K*	2382	3673	1703
Steady state design fuel temperature limit/*K*	2473	2473	2473
Steady state design cladding temperature limit/*K*	1873	2533	1073
Fuel temperature limit under accident/*K*	2773	2773	2773
cladding temperature limit under accident/ *K*	2073 (LOCA) 1873 (LOFA)	2073 (LOCA) 1873 (LOFA)	1473 (LOCA) 1273 (LOFA)
Pressure limit of containment	–	–	1.2 times

3.3. System model and validation

According to the physical property calculation model, heat transfer model and turbomachinery calculation model proposed above, the function of SCTRAN code is expanded and improved, and the development of simulation calculation code for He–Xe mixture gas reactor system has been completed. And based on the diagram of a CBC loop in S^4 reactor shown in Fig. 10, this paper divides the nodes of three CBC loop in S^4 reactor. Fig. 12 shows the node diagram of one of the three CBC loops. It should be noted that boundary conditions are used to present the cold side of the gas cooler, as the heat pipe radiator model has not been developed in SCTRAN. Comparison between calculated and design values of system steady-state are shown in Table 8. It can be found that the accuracy of the system model is validated with original design parameters of S^4 reactor system (Gallo and El-Genk, 2009), with the maximum error is within 0.5 %.

4. In-orbit safety performance evaluation of S^4 reactor

During the He-Xe cooled reactor on-orbit operation, it is essential to ensure that it meets various safety objectives under different operational accidents. Due to the high-temperature operation design of the He-Xe cooled reactor, it is required that the core does not melt under accidents. The advanced design report for space reactors (Burdge et al., 2006) indicates that the alternator power must not exceed 110 % of its rated capacity. The Parasitic Load Radiator (PLR) is a thermal management system designed to dissipate excess power in conjunction with a speed regulator. It employs Nichrome V resistive elements operating at 1255 K, insulated by beryllium oxide, and a carbon–carbon radiator plate for high-temperature stability and structural integrity (Metcalf, 1993). And the Parasitic Load Radiator (PLR) can absorb the output of the alternator to accommodate speed regulation at full power conditions. The rated power of the PLR is generally the same as that of the alternator. Based on typical operating events and safety design limits of space nuclear power reactor (safety design limits of some typical space nuclear power reactor are shown in Table 9), this paper studies the response characteristics of the S^4 reactor Brayton cycle system under reactivity insertion and mechanical failure of Brayton loop in detail,

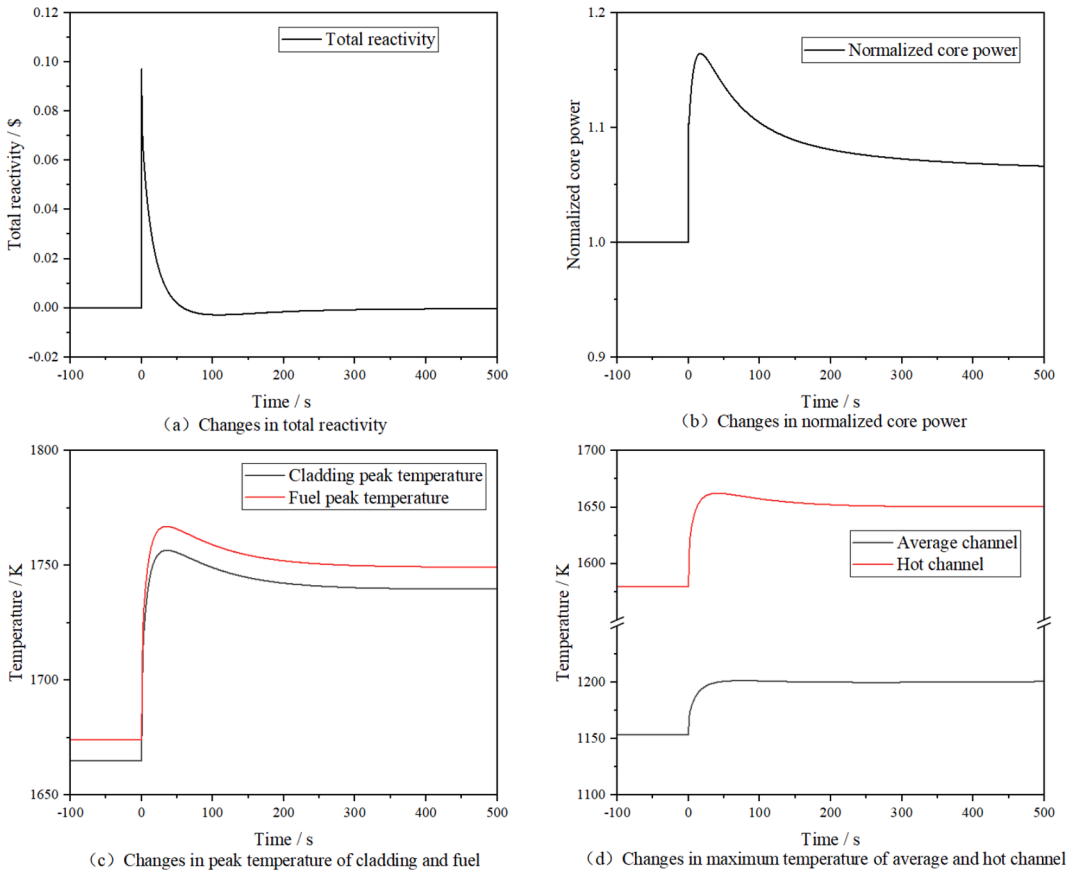


Fig. 13. Variation of core parameters over time during a step insertion of $ 0.1 positive reactivity.

which are typical accident of space reactor.

4.1. Reactivity insertion accident

When the S^4 reactor operates in orbit, the unexpected rotation of the control drum may insert positive reactivity into the reactor core. Assume that the shaft speed of the turbomachinery remains constant (Zhang et al., 2021) (Burdge et al., 2006). This can cause the coolant and fuel temperatures to rise, leading to an increase in the output power of the rotational shaft. Thus, to obtain the transient response of the S^4 reactor during a positive reactivity insertion accident, a simulation of the unexpected rotation of the control drum without triggering the protection system is conducted.

4.2. Step positive reactivity insertion

The S^4-CBC space reactor power system operates in a steady-state regime for an extended period, ensuring stable thermal and power output under nominal conditions. At the onset of the accident, $ 0.1 positive reactivity is inserted to the core by the unexpected rotation of the control drum at 0 s (the maximum total positive reactivity that can be inserted by the unexpected rotation of single control drum is $ 0.68 (King and El-Genk, 2009). The changes in thermal and hydraulic parameters of the core and Brayton cycle system are analyzed as the following.

Fig. 13(a) shows the variation of total reactivity over time during a step insertion of $ 0.1 positive reactivity. It can be observed that the total reactivity reaches a peak value of $ 0.088 instantaneously upon the step reactivity insertion. Due to the negative temperature reactivity feedback of the core structural materials and coolant, the increase in coolant and fuel temperatures introduces negative reactivity into the core. The total reactivity decreases to $ 0 within 300 s and returns to a critical stable state.

As can be seen from the variation curve of the normalized core power in Fig. 13(b), the step positive reactivity insertion accident caused by the mis-operation of the control drum makes the core thermal power increase rapidly to 116.4 % of the rated power. Then, under the effect of core negative feedback, it finally drops to a new stable power, which is 105.6 % of the rated power, within 500 s.

Fig. 13(c) and (d) illustrate the changes in the maximum temperature of the cladding, fuel and coolant during the accident. Due to the increase in core power, the temperature of the fuel and cladding rise rapidly. The maximum temperature of fuel and cladding increase to 1767 K and 1756.6 K within 35 s, which are 1356 K lower than the melting point of UN. When the system resumes stable operation at 500 s, the maximum temperatures of fuel and cladding are 1739.9 K and 1749.5 K, respectively. Given the small specific heat capacity of the He – Xe mixture gas, the coolant channel temperature rises rapidly following the increase in the cladding temperature. The maximum coolant temperature at the hot channel and the average channel increase to 1662.3 K and 1201.3 K within 40 s.

As shown in Fig. 14(a), the coolant mass flow rate decreases instantaneously from 1.8076 kg/s to 1.8058 kg/s. Because the increase in coolant temperature leads to thermal expansion, which results in a decrease in density. Concurrently, the viscosity of the coolant increases with the rising temperature, which raises the flow resistance. Once the coolant temperature begins to decrease, the mass flow rate starts to rise and eventually stabilizes around 500 s.

Fig. 14(b) and (d) show the changes in inlet and outlet temperature and normalized power of turbine and compressor. Due to the increase in core outlet temperature, the inlet and outlet temperatures of the turbine increased from 1154.6 K and 1018.1 K to 1200 K and 1059.1 K, resulting in an increase in turbine power to 103.8 % of rated power (as shown in Fig. 14(d)). The inlet temperature of the compressor remains almost

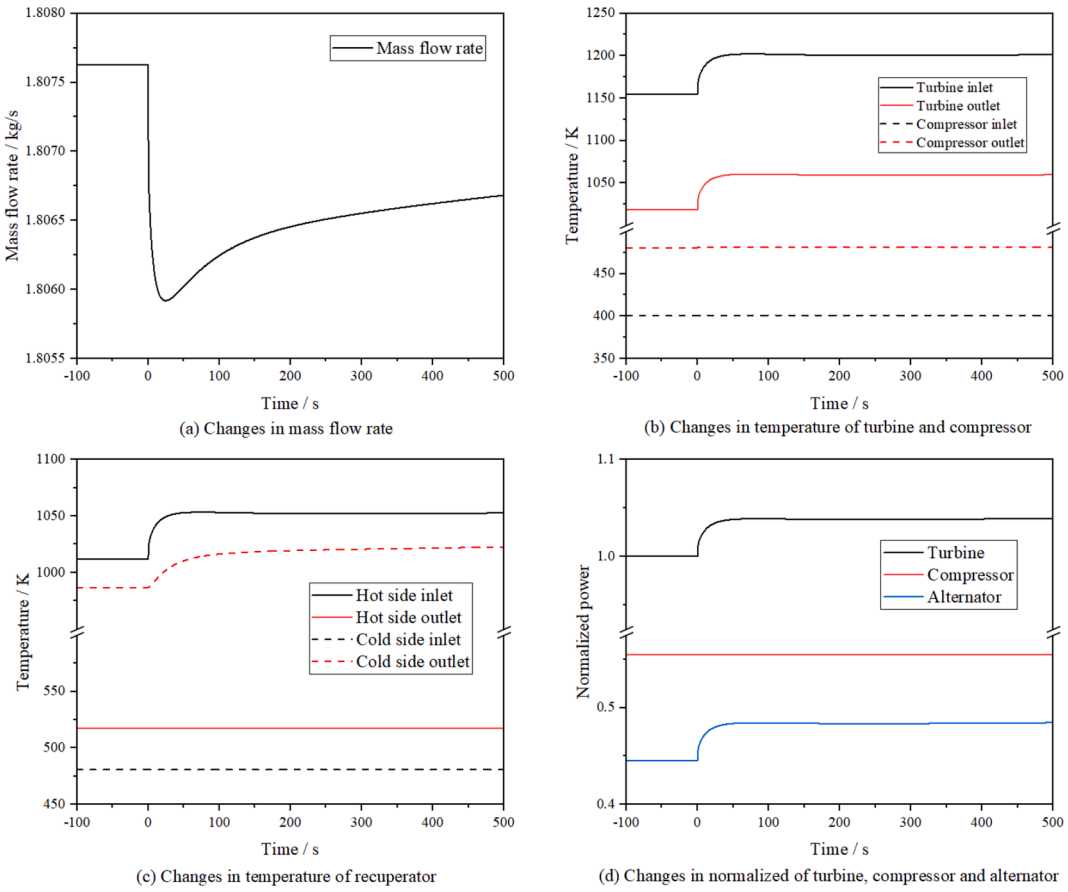


Fig. 14. Variation of system parameters over time during a step insertion of $ 0.1 positive reactivity.

constant due to the strong cooling capacity of the gas cooler. However, the slight decrease in the mass flow rate leads to a slight decrease in the compressor power. The load increases to 108.8 % of the rated power because of keeping the speed of the shaft, which is less than 110 % of its rated capacity.

Fig. 14(c) shows the variation of inlet and outlet temperatures on the cold and hot sides of the recuperator over time. The temperature of the He-Xe at the inlet of the hot side rises due to the increase in outlet temperature of the turbine. The inlet temperature of the hot side increases from 1012.2 K to 1052.3 K, while the outlet temperature of the hot side remains almost constant because of the strong heat transfer capacity of the recuperator. The cold side inlet temperature is basically unchanged (the compressor outlet temperature is basically unchanged), and the cold side outlet temperature increases from 986.8 K to 1022.1 K.

4.3. *Sensitivity analysis of step reactivity insertion*

The S^4-CBC space reactor power system is still at the conceptual design stage, and several transient parameters remain undetermined. Based on the maximum total positive reactivity of $ 0.68 that can be inserted by the unexpected rotation of a single control drum, $0.68 is selected as the worst-case simulation result. To better observe the transient response of the system as the magnitude of reactivity insertion increases linearly, $ 0.2 and $ 0.4 are selected within the range of $ 0.1 to $ 0.68. Thus, this paper centers on the response characteristics of the Brayton system under step positive reactivity insertions of $ 0.1, $ 0.2, $ 0.4, and $ 0.68.

As shown in Fig. 15(a), it can be seen that the peak power of the core increases with the step reactivity insertion, and the rate of this increase gets faster and faster. Similarly, the steady-state power of the core also rises with the step reactivity insertion, and the rate remains constant. When a step positive reactivity insertion of $ 0.68 occurs, the peak power reaches 251.5 % of the rated power. Therefore, the S^4-CBC space reactor power system should avoid large-scale step positive reactivity insertion accidents.

The transient and steady-state peak temperatures of the cladding and fuel increase linearly as the step reactivity is inserted. The difference between the peak temperature and the steady-state temperature of the cladding and fuel also rises with the step reactivity insertion, as depicted in Fig. 15(b). The minimum and steady-state mass flow rates decline, and the difference between the minimum and steady-state mass flow rates grows with the step reactivity insertion, as shown in Fig. 15(c). The maximum pressure increases linearly with the step reactivity insertion, as presented in Fig. 15(d). During the step positive reactivity insertion of $ 0.1, the inlet pressure of the core rises to 101.5 % of the rated pressure. When the step positive reactivity insertion is $ 0.68, the inlet pressure of the core climbs to 109.6 % of the rated pressure.

As mentioned above, the shaft speed of the turbomachinery remains constant, and the maximum alternator power is 110 % of the rated capacity. Fig. 16 shows the maximum core temperature and alternator power during different positive reactivity insertions. When the introduced reactivity does not exceed $ 0.11, the S^4-CBC space reactor power system can keep the power and temperature stable and below the safety limits by itself negative reactivity feedback. Moreover, additional shaft power control measures are necessary to maintain the shaft speed when the reactivity insertion exceeds $ 0.11. With the inserted reactivity of $ 1.32, the maximum core temperature will reach the fuel melting point (3123 K), which will cause core melting. If the S^4-CBC space reactor power system has a Parasitic Load Radiator (PLR), the load limit can be increased to 210 % of the rated power, and the reactivity that the system can endure will increase to $ 1.08.

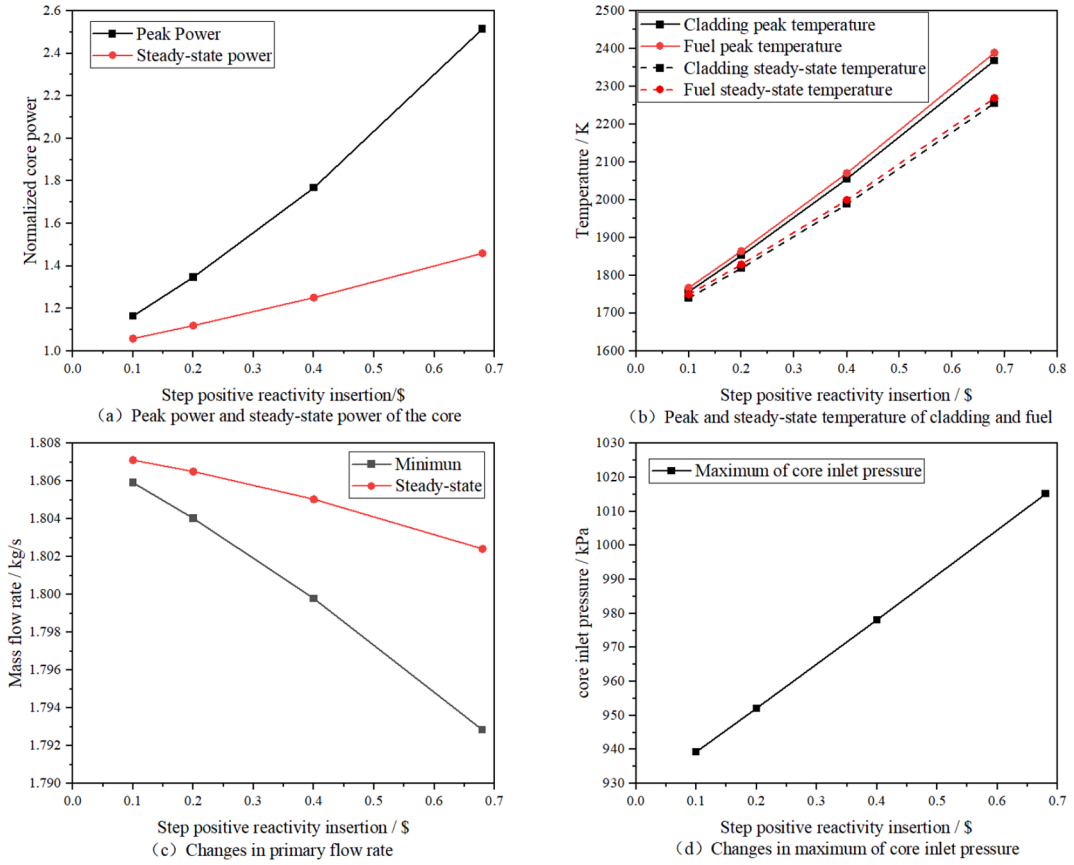


Fig. 15. Variation of core parameters over time during different positive reactivity insertion.

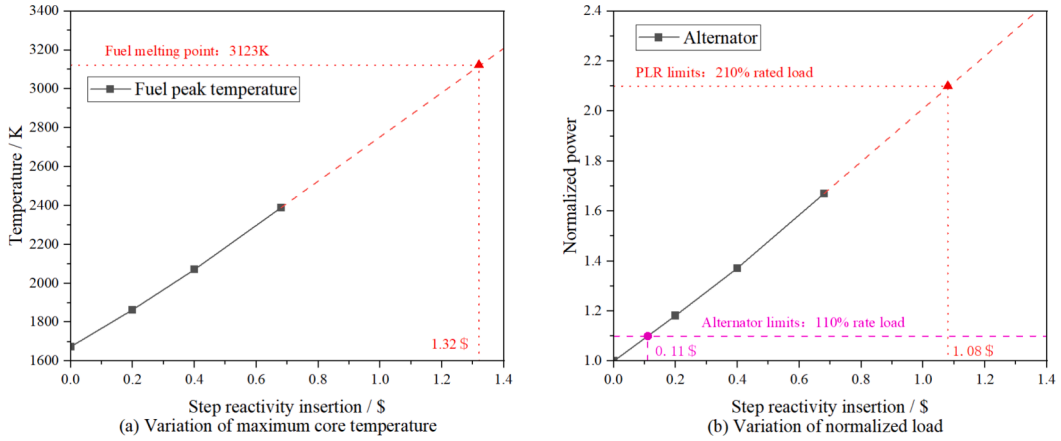


Fig. 16. Variation of maximum core temperature and normalized load during different positive reactivity insertion.

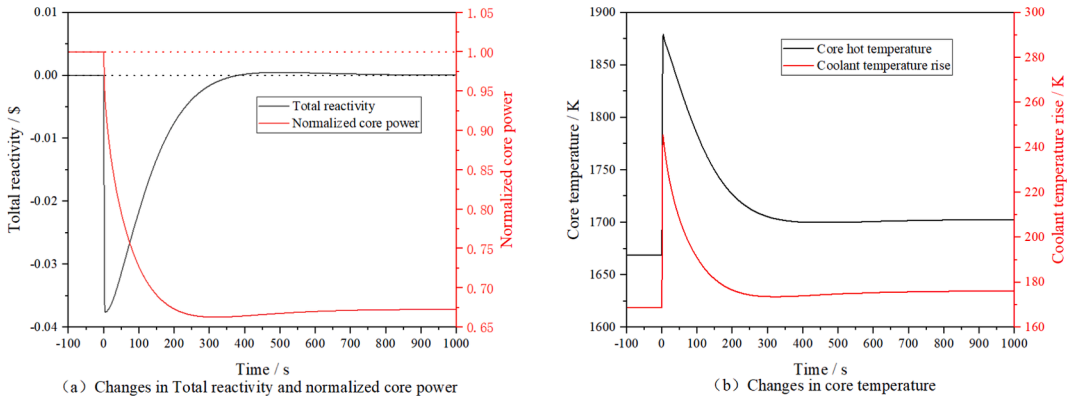
4.4. Mechanical failure of Brayton loop

The power conversion unit in the S^4-CBC space reactor power system has three independent Brayton systems. During the in-orbit operation, each Brayton system operates at one-third load. The main purpose of the design is to increase the redundancy of the power conversion system and improve the reliability of the power supply. This section analyzes the system response characteristics during mechanical failure of one and two Brayton loop (MFOBL1 and MFOBL2). It is assumed that turbomachinery of one or two Brayton loops shutdown during the accident, resulting in an instantaneous decrease coolant mass flow rate in core.

4.5. Mechanical failure of one Brayton loop

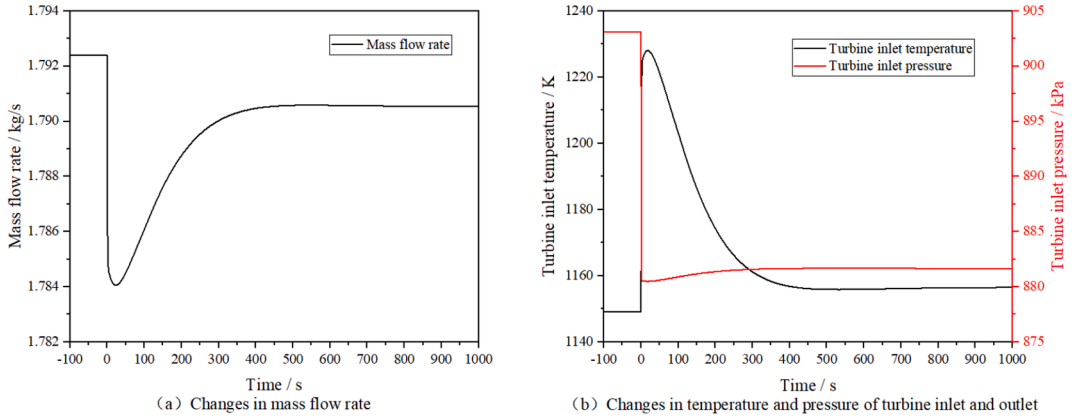
Mechanical failure of one loop occurs (rotor jamming), and the BL2 and BL3 work normally after the S^4-CBC space reactor power system operates in a steady-state regime for an extended period, ensuring stable thermal and power output under nominal conditions. At the beginning of the accident, the mass flow rate of the failure loop is decreased to 0 kg/s within 0.1 s. The changes in thermal and hydraulic parameters of the core and Brayton cycle system are analyzed as the following.

The mass flow rate in core instantly decreases to two-thirds of its original value during the accident. Fig. 17 shows the variation of core parameters over time during MFOBL1. At the rated power of the core, the temperature of the fuel and cladding rapidly increases (Fig. 17 (b)),



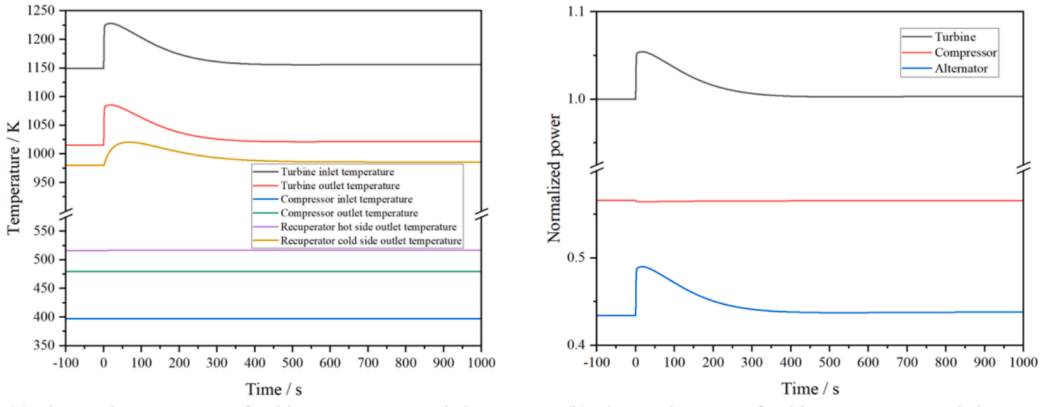
(a) Changes in Total reactivity and normalized core power (b) Changes in core temperature

Fig. 17. Variation of core parameters over time during MFOBL1.



(a) Changes in mass flow rate (b) Changes in temperature and pressure of turbine inlet and outlet

Fig. 18. Variation of the BL2 turbine inlet parameters over time during MFOBL1.



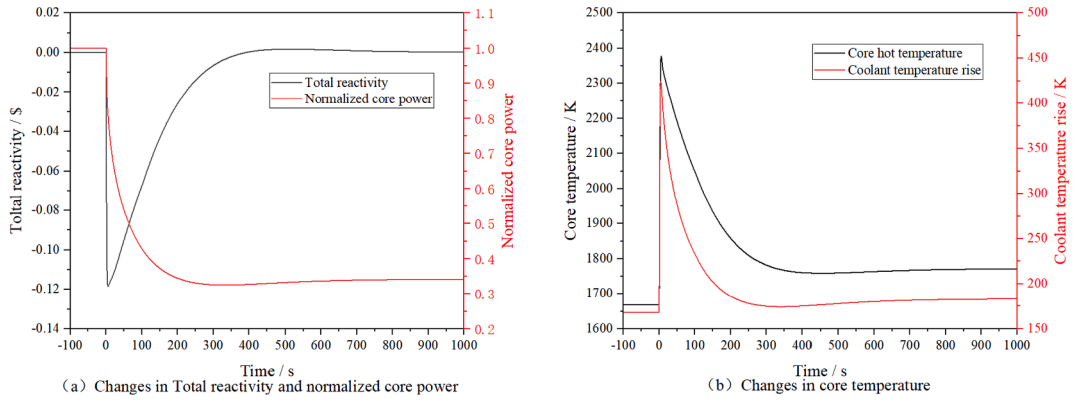
(a) Changes in temperature of turbine, compressor and alternator (b) Changes in power of turbine, compressor and alternator

Fig. 19. Variation of the BL2 system parameters over time during MFOBL1.

and the decrease in core power is caused by negative reactivity feedback (Fig. 17 (a)). The hot spot temperature reaches its peak of 1878 K (1245 K lower than the melting point) within 5 s, and begins to decrease due to a decrease in power. Finally, under the effect of negative reactivity feedback, the total reactivity decreased to $ 0 at around 800 s, and the core power stabilized at 67.3 % of the rated power. The rise of core inlet and outlet coolant temperature increases from 168 K to 176 K.

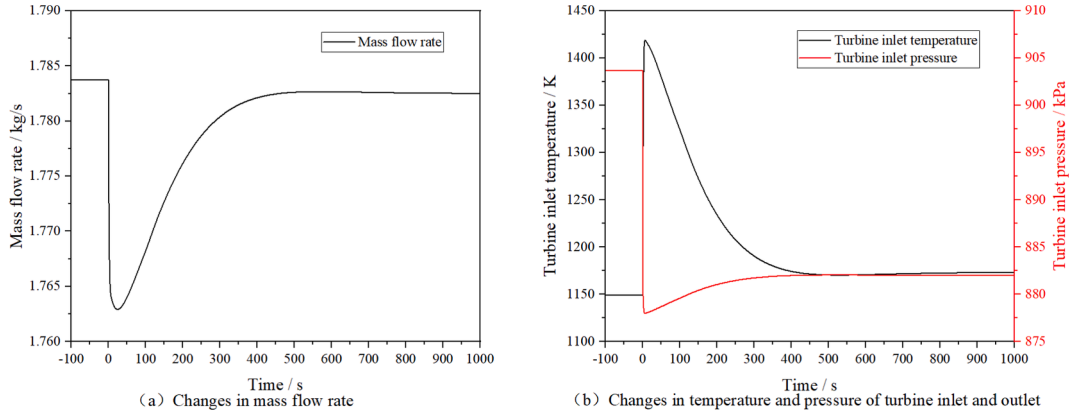
Fig. 18 shows the variation of the BL2 turbine inlet parameters over time during MFOBL1. The mass flow rate is proportional to the turbine inlet pressure and inversely proportional to the inlet temperature. When the MFOBL1 occurs, the inlet temperature of the BL2 turbine increases and the inlet pressure decreases (as shown in Fig. 18(b)). Therefore, the mass flow rate decreased from 1.7925 kg/s to 1.784 kg/s. Then it gradually increased as the inlet temperature decreased, eventually stabilizing at 1.791 kg/s (as shown in Fig. 18(a)).

Fig. 19 shows the variation of BL2 system parameters over time during MFOBL1. The inlet temperature of the turbine increases from 1149.2 K to 1228 K, and it finally stabilizes at 1156.4 K, while the inlet and outlet temperatures of the compressor remain almost constant. Therefore, the peak power of the turbine is 105.4 % of the rated power, and the compressor power is slightly reduced. Due to the constant shaft speed, the peak load is 112.8 % of the rated value, which is 112.8 % of



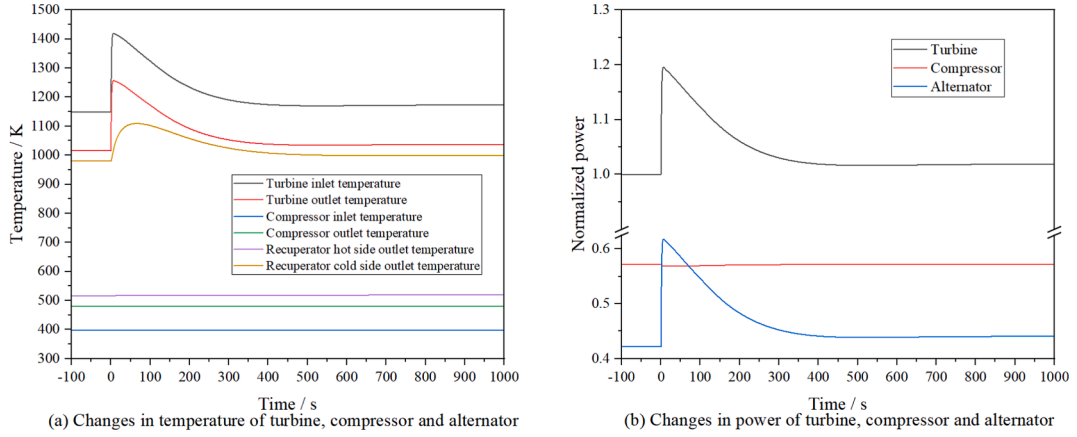
（a）Changes in Total reactivity and normalized core power （b）Changes in core temperature

Fig. 20. Variation of core parameters over time during MFOBL2.



（a）Changes in mass flow rate （b）Changes in temperature and pressure of turbine inlet and outlet

Fig. 21. Variation of the BL3 turbine inlet parameters over time during MFOBL2.



(a) Changes in temperature of turbine, compressor and alternator (b) Changes in power of turbine, compressor and alternator

Fig. 22. Variation of the BL3 system parameters over time during MFOBL2.

the rated value, exceeding the rated value by 2.8 %. Therefore, it is necessary to transmit the excess power to the Parasitic Load Radiator or control the shaft power.

4.6. *Mechanical failure of two Brayton loops*

Mechanical failure of two loops (MFOBL2) occurs (rotor jamming), and the BL3 works normally after the S^4-CBC space reactor power system operates in a steady-state regime for an extended period, ensuring stable thermal and power output under nominal conditions. At the beginning of the accident, the mass flow rate of the two failure loops is decreased to 0 kg/s within 0.1 s. The changes in thermal and hydraulic parameters of the core and Brayton cycle system are analyzed as the following.

When the MFOBL2 (mechanical failure of two loops) occurs, the mass flow rate in the core drops instantaneously to one-third of its original value. As depicted in Fig. 20, at the core's rated power, the temperatures of the fuel and cladding shoot up rapidly (Fig. 20(b)). Meanwhile, the core power decline is triggered by negative reactivity feedback, as shown in Fig. 20(a). The hot spot temperature peaks at 2377 K, which is 746 K lower than the melting point, within just 5 s. Subsequently, it begins to fall as the power reduces. Eventually, under the influence of negative reactivity feedback, the total reactivity reaches

$ 0 around 800 s, and the core power stabilizes at 34.2 % of the rated power. The temperature rise of the core inlet and outlet coolant climbs from 168 K to 183 K.

Fig. 21 and Fig. 22 illustrate the variation over time of the BL3 turbine inlet parameters and system parameters during the MFOBL2 event. The pattern of parameter changes is the same as that in the MFOBL1 case. Specifically, the peak power of the turbine amounts to 119.5 % of the rated power. Due to the constant shaft speed, the peak load reaches 146.5 % of the rated value. Hence, it is essential to transfer the excess power to the Parasitic Load Radiator (PLR) or regulate the shaft power.

5. Conclusion

This paper develops a new system code for closed Brayton cycle of the He-Xe cooled space reactor. Based on the present system analysis code SCTRAN, the physical property, flow resistance and heat transfer correlation, as well as turbomachinery models for He-Xe are further developed. Then, the safety characteristics of the S^4-CBC space reactor power system during the typical accidents are investigated. The main conclusions are summarized as follows:

The calculated values from the developed property calculation code closely align with the standard values. Blasius and four heat transfer correlations are evaluated based on experimental data, which proves that the Blasius and Taylor correlations can well predict flow and heat transfer calculations for He-Xe. The maximum prediction error of the proposed turbomachinery models is within 2 %. Compared with the reference design values, the core and system model calculation results have maximum relative errors of 3.43 % and 0.47 % respectively, which proves that the code is reliable for simulating the thermal–hydraulic behavior of the S^4-CBC space reactor power system.

During the reactivity insertion accident, the S^4-CBC space reactor power system can keep the power and temperature stable and below the safety limits by itself negative reactivity feedback with the reactivity insertion below $ 0.11. If the S^4-CBC space reactor power system applies the Parasitic Load Radiator, the reactivity insertion that the system can endure will increase to $ 1.08 enhancing its safety margin during reactivity insertion accident. Otherwise, additional shaft power control measure is necessary to maintain the shaft speed when the reactivity insertion exceeds $ 0.11. With the reactivity insertion of $ 1.32, core temperature control measure is needed because the maximum core temperature will reach the fuel melting point (3123 K). During the mechanical failures of one and two Brayton loops, the maximum fuel temperature has a safety margin of 1245 K and 746 K from the melting point, suggesting that the reactor has a favorable performance to endure this type of accident. However, it is necessary to control the shaft power or output the excess load to the Parasitic Load Radiator.

In subsequent research, further consideration should be given to the design of control strategies for the S4-CBC space reactor power system to ensure that, in the event of accidents, the core temperature, shaft speed, and load do not exceed their respective safety limits.

CRediT authorship contribution statement

Ming Liu: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Pan Wu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Ke Huang:** Supervision, Project administration, Methodology, Investigation, Conceptualization. **Jianqiang Shan:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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